Topology I

Lecture notes, Chapters 1–8 (General Topology)

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Chapter 1. Set Theory and Logic

This chapter collects the set theory used in the rest of the course. Most of it is routine; the essential new tools are countability arguments, the axiom of choice, and its equivalent forms (well-ordering and Zorn's lemma).

1.1 Sets, functions and relations

We use the usual operations $A\cup B$, $A\cap B$, $A-B$, products $A\times B$, and De Morgan's laws $X-\bigcup A_\alpha = \bigcap(X-A_\alpha)$, $X-\bigcap A_\alpha = \bigcup(X-A_\alpha)$. A **function** $f:A\to B$ is a rule (formally a subset of $A\times B$) assigning to each $a\in A$ one $f(a)\in B$. It is **injective** if $f(a)=f(a')\Rightarrow a=a'$, **surjective** if $f(A)=B$, and **bijective** if both. Images and preimages satisfy

$$f^{-1}\Big(\bigcup B_\alpha\Big)=\bigcup f^{-1}(B_\alpha),\quad f^{-1}\Big(\bigcap B_\alpha\Big)=\bigcap f^{-1}(B_\alpha),\quad f^{-1}(B-C)=f^{-1}(B)-f^{-1}(C),$$

whereas images preserve unions but only $f(\bigcap A_\alpha)\subset\bigcap f(A_\alpha)$ in general (equality for injective f). Also $f(f^{-1}(B))\subset B$ and $A\subset f^{-1}(f(A))$.

A **relation** on A is a subset $C\subset A\times A$. An **equivalence relation** is reflexive, symmetric and transitive; its equivalence classes partition A, and conversely every partition comes from one. An **order relation** (simple or linear order) $<$ is comparable ($x\neq y\Rightarrow x<y$ or $y<x$), nonreflexive and transitive. Two ordered sets have the same **order type** if there is an order-preserving bijection between them. For subsets of an ordered set we have upper bounds, suprema and infima, and the **least upper bound property**: every nonempty subset bounded above has a supremum. $\mathbb{R}$ is characterized as an ordered field with this property; $\mathbb{Z}_+$ satisfies the **well-ordering property** (every nonempty subset has a least element), which is equivalent to induction.

1.2 Finite, countable and uncountable sets

A set is **finite** if it is empty or in bijection with $\{1,\dots,n\}$ for some n; this n (the cardinality) is unique, since there is no bijection of $\{1,\dots,n\}$ with a proper subset of itself (pigeonhole principle, by induction on n). A set is **countably infinite** if it is in bijection with $\mathbb{Z}_+$, and **countable** if finite or countably infinite.

Theorem 1.1

For nonempty B the following are equivalent: (a) B is countable; (b) there is a surjection $\mathbb{Z}_+\to B$; (c) there is an injection $B\to\mathbb{Z}_+$. Subsets of countable sets are countable; countable unions and finite products of countable sets are countable.

Proof. (c)⇒(a): an infinite subset $C\subset\mathbb{Z}_+$ is enumerated by $h(n)=$ the smallest element of $C-h(\{1,\dots,n-1\})$ (recursive definition, Section 1.3). (b)⇒(c): send b to the least element of its preimage. $\mathbb{Z}_+\times\mathbb{Z}_+\to\mathbb{Z}_+$, $(n,m)\mapsto 2^n3^m$ is injective. A countable union $\bigcup_n A_n$ is the image of $\mathbb{Z}_+\times\mathbb{Z}_+$ under $(n,m)\mapsto f_n(m)$, where $f_n:\mathbb{Z}_+\to A_n$ are surjections (chosen using the axiom of choice). ■

Theorem 1.2

$\{0,1\}^\omega$ (infinite binary sequences) and $\mathcal{P}(\mathbb{Z}_+)$ are uncountable. More generally, there is no surjection $A\to\mathcal{P}(A)$.

Proof. Given $g:A\to\mathcal{P}(A)$, the set $B=\{a:a\notin g(a)\}$ is not in the image: if $B=g(a_0)$, then $a_0\in B\iff a_0\notin B$. For sequences, the diagonal sequence $y_n=1-x_n(n)$ differs from every x_n. ■

So there are infinitely many sizes of infinite sets. The **Schröder–Bernstein theorem** states that injections $A \to B$ and $B \to A$ give a bijection; in particular $\mathbb{R}$, $\mathcal{P}(\mathbb{Z}_+)$ and $\{0,1\}^\omega$ have the same cardinality.

1.3 Recursive definition and the axiom of choice

Theorem 1.3 (Principle of recursive definition)

Given a set A, an element $a_1 \in A$, and a rule ρ assigning to each function $f : \{1,\dots,n\} \to A$ an element $\rho(f) \in A$, there is a unique $h : \mathbb{Z}_+ \to A$ with $h(1) = a_1$ and $h(n) = \rho(h|_{\{1,\dots,n-1\}})$ for $n > 1$.

Proof. Show by induction that for each n there is a unique function on $\{1,\dots,n\}$ satisfying the formula there; these functions are compatible, and their union is h. ∎

Definition (Axiom of choice)

For any collection $\mathcal{A}$ of disjoint nonempty sets there is a set containing exactly one element of each member of $\mathcal{A}$. Equivalently, for any collection $\mathcal{B}$ of nonempty sets there is a **choice function** $c : \mathcal{B} \to \bigcup \mathcal{B}$ with $c(B) \in B$.

Theorem 1.4

A set A is infinite iff there is an injection $\mathbb{Z}_+ \to A$ iff A is in bijection with a proper subset of itself.

Proof. If A is infinite, choose (with a choice function c on nonempty subsets of A) recursively $f(n) = c(A - f(\{1,\dots,n-1\}))$. Given an injection f, the map shifting $f(n) \mapsto f(n+1)$ and fixing the rest is a bijection onto a proper subset. A finite set has no such bijection. ∎

1.4 Well-ordered sets and Zorn's lemma

A set is **well-ordered** if it is simply ordered and every nonempty subset has a least element. Each element α of a well-ordered set J has a **section** $S_\alpha = \{x : x < \alpha\}$.

Theorem 1.5 (Well-ordering theorem, Zermelo)

Every set can be well-ordered. (This is equivalent to the axiom of choice.)

Proposition 1.6

There is a well-ordered set S_Ω that is uncountable but every section of which is countable; it is unique up to order type and is called the **minimal uncountable well-ordered set**. Every countable subset of S_Ω has an upper bound in S_Ω.

Proof. Well-order an uncountable set B, append a largest element, and let Ω be the least element whose section is uncountable; then S_Ω works. A countable subset A has $\bigcup_{a\in A} S_a$ countable, hence not all of S_Ω; any element outside is an upper bound. ∎

S_Ω and $\overline{S}_\Omega = S_\Omega \cup \{\Omega\}$ give many counterexamples in later chapters.

A **strict partial order** $\prec$ is nonreflexive and transitive (not every pair need be comparable).

Theorem 1.7 (Zorn's lemma)

Let A be partially ordered. If every simply ordered subset (chain) of A has an upper bound in A, then A has a maximal element.

Idea. Using a choice function, build by transfinite recursion along a well-ordering a chain that can no longer be extended; its upper bound is maximal. (Equivalently, the Hausdorff maximum principle: every partially ordered set has a maximal chain.) ■

Example 1.1

Zorn's lemma shows that every vector space has a basis (a maximal linearly independent set spans), that every ring with identity has a maximal ideal, and (Chapter 5) it is the engine behind the Tychonoff theorem.

Chapter 2. Topological Spaces and Continuous Functions

2.1 Topological spaces and bases

Definition (Topology)

A **topology** on a set X is a collection $\mathcal{T}$ of subsets (the **open sets**) containing $\emptyset$ and X and closed under arbitrary unions and finite intersections.

Examples: the discrete topology (all subsets), the indiscrete topology $\{\emptyset, X\}$, the finite complement topology (U open iff $X - U$ is finite or $U = \emptyset$). $\mathcal{T}'$ is **finer** than $\mathcal{T}$ if $\mathcal{T} \subset \mathcal{T}'$.

Definition (Basis)

A **basis** is a collection $\mathcal{B}$ of subsets such that every x lies in some $B \in \mathcal{B}$, and if $x \in B_1 \cap B_2$ there is $B_3 \in \mathcal{B}$ with $x \in B_3 \subset B_1 \cap B_2$. The topology it generates consists of all unions of elements of $\mathcal{B}$. A **subbasis** is any collection whose union is X; finite intersections of its members form a basis.

Lemma 2.1

(a) If $\mathcal{C}$ is a collection of open sets such that every open U and $x \in U$ admit $C \in \mathcal{C}$ with $x \in C \subset U$, then $\mathcal{C}$ is a basis for the topology. (b) The topology of $\mathcal{B}'$ is finer than that of $\mathcal{B}$ iff for every $B \in \mathcal{B}$ and $x \in B$ there is $B' \in \mathcal{B}'$ with $x \in B' \subset B$.

On $\mathbb{R}$: the **standard topology** has basis the open intervals; the **lower limit topology** $\mathbb{R}_\ell$ has basis $[a, b)$; the **K-topology** $\mathbb{R}_K$ has basis (a, b) and $(a, b) - K$, $K = \{1/n\}$. Both $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are strictly finer than the standard topology and are not comparable with each other.

2.2 Order, product and subspace topologies

For a simply ordered set X with more than one element, the **order topology** has basis the open intervals (a, b) together with $[a_0, b)$ or $(a, b_0]$ if X has a least element a_0 or largest element b_0. The open rays form a subbasis. Examples: $\mathbb{R}$; $\mathbb{Z}_+$ (discrete); the dictionary order on $\mathbb{R} \times \mathbb{R}$; $\{1, 2\} \times \mathbb{Z}_+$ (not discrete: 2×1 has no smallest open neighbourhood avoiding $1 \times \mathbb{Z}_+$).

The **product topology** on $X \times Y$ has basis $U \times V$ with U, V open; if $\mathcal{B}, \mathcal{C}$ are bases, $\{B \times C\}$ is a basis. Equivalently it has subbasis $\pi_1^{-1}(U)$, $\pi_2^{-1}(V)$.

For $Y \subset X$, the **subspace topology** is $\{Y \cap U : U \text{ open in } X\}$. If Y is open in X, open sets of Y are open in X. The product of subspaces is the subspace of the product. But order and subspace topologies can differ: on $Y = [0, 1) \cup \{2\} \subset \mathbb{R}$ the point $\{2\}$ is open in the subspace topology but not in the order topology of Y. They agree when Y is convex (an interval) in X.

2.3 Closed sets and limit points

A is **closed** if $X - A$ is open; closed sets are closed under arbitrary intersections and finite unions. The **closure** $\overline{A}$ is the intersection of all closed sets containing A, the **interior** $\operatorname{Int} A$ the union of all open sets inside A.

Theorem 2.2

$x \in \overline{A}$ iff every open set containing x (equivalently every basis element containing x) meets A. If A' is the set of limit points of A (x such that every neighbourhood of x meets $A - \{x\}$), then $\overline{A} = A \cup A'$; A is closed iff $A' \subset A$. In a subspace Y, the closure of $A \subset Y$ is $\overline{A} \cap Y$.

Definition (Hausdorff space)

X is **Hausdorff** if distinct points have disjoint neighbourhoods. X satisfies T_1 if finite sets are closed.

In a Hausdorff space finite sets are closed, a sequence converges to at most one point, and in a T_1 space $x \in A'$ iff every neighbourhood of x contains infinitely many points of A. Order topologies, products and subspaces of Hausdorff spaces are Hausdorff. The finite complement topology on $\mathbb{R}$ is T_1 but not Hausdorff, and there the sequence $1, 2, 3, \ldots$ converges to every point.

2.4 Continuous functions

Definition (Continuity, homeomorphism)

$f : X \to Y$ is **continuous** if $f^{-1}(V)$ is open for every open V. A continuous bijection with continuous inverse is a **homeomorphism**; an injective continuous map that is a homeomorphism onto its image is an **imbedding**.

Theorem 2.3

The following are equivalent for $f : X \to Y$: (a) f is continuous; (b) $f(\overline{A}) \subset \overline{f(A)}$ for all A; (c) preimages of closed sets are closed; (d) for each x and each neighbourhood V of $f(x)$ there is a neighbourhood U of x with $f(U) \subset V$.

Constructions: constants, inclusions, compositions, restrictions, and maps that are continuous on each set of an open cover are continuous.

Theorem 2.4 (Pasting lemma)

If $X = A \cup B$ with A, B closed and $f : A \to Y$, $g : B \to Y$ are continuous and agree on $A \cap B$, then the combined function is continuous.

Proof. For closed C, $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$ is a union of two sets closed in A, B, hence in X. ■

$f : A \to X \times Y$ is continuous iff both coordinates are.

2.5 The product topology

For an indexed family $\{X_\alpha\}_{\alpha \in J}$ the **box topology** on $\prod X_\alpha$ has basis $\prod U_\alpha$ with all U_α open, and the **product topology** has basis $\prod U_\alpha$ with $U_\alpha = X_\alpha$ for all but finitely many α (subbasis $\pi_\beta^{-1}(U_\beta)$). They agree for finite products.

Theorem 2.5

With the product topology, $f : A \to \prod X_\alpha$ is continuous iff each $\pi_\alpha \circ f$ is continuous. This fails for the box topology.

Proof. Preimages of subbasis elements are $(\pi_\beta \circ f)^{-1}(U_\beta)$. For the box topology on $\mathbb{R}^\omega$, $f(t) = (t, t, t, \dots)$ has continuous coordinates, but $f^{-1}(\prod(-\frac{1}{n}, \frac{1}{n})) = \{0\}$ is not open. ■

Products and subspaces commute, products of Hausdorff spaces are Hausdorff, and $\prod \overline{A_\alpha} = \overline{\prod A_\alpha}$ in either topology. Unless stated otherwise, products carry the product topology.

2.6 Metric topologies

A metric d gives the topology with basis the ϵ-balls $B_d(x, \epsilon)$; X is **metrizable** if some metric induces its topology. Examples: the Euclidean and square metrics on $\mathbb{R}^n$ give the product topology; the **uniform metric** $\bar{\rho}(x, y) = \sup \min(|x_\alpha - y_\alpha|, 1)$ on $\mathbb{R}^J$ gives a topology between product and box.

Theorem 2.6

$\mathbb{R}^\omega$ with the product topology is metrizable, by $D(x, y) = \sup_n \frac{\min(|x_n - y_n|, 1)}{n}$. Countable products of metrizable spaces are metrizable.

Proof. A D-ball of radius ϵ around x contains the product basis set $\prod_{n \le N}(x_n - \epsilon, x_n + \epsilon) \times \mathbb{R} \times \cdots$ with $1/N < \epsilon$, and conversely a product basis set around x restricting finitely many coordinates contains a small D-ball. ■

In metric spaces continuity is the ϵ–δ condition, and the **sequence lemma** holds: $x \in \overline{A}$ iff a sequence in A converges to x (the "if" part in any space; the "only if" needs a countable basis at x). Hence f is continuous iff it preserves convergent sequences. A uniform limit of continuous maps into a metric space is continuous. These facts show that $\mathbb{R}^\omega$ in the box topology and $\mathbb{R}^J$ (J uncountable) in the product topology are not metrizable.

2.7 The quotient topology

A surjective continuous $p : X \to Y$ is a **quotient map** if $U \subset Y$ is open whenever $p^{-1}(U)$ is open. Open or closed continuous surjections are quotient maps. For a partition X^* of X, the **quotient space** carries the finest topology making the projection continuous.

Theorem 2.7

Let $p : X \to Y$ be a quotient map and $g : X \to Z$ constant on each fibre $p^{-1}(y)$. Then $g = f \circ p$ for a unique $f : Y \to Z$, and f is continuous; f is a quotient map iff g is.

Example 2.1

(a) The square $[0, 1]^2$ with opposite edges identified is the torus; with all boundary points identified to one point it is the sphere S^2.
(b) Products of quotient maps need not be quotient maps; and quotients of Hausdorff spaces need not be Hausdorff (collapse $\mathbb{Q}$ in $\mathbb{R}$).
(c) $p : \mathbb{R} \to \mathbb{R}/\mathbb{Z} \cong S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$, is a quotient map.

Chapter 3. Connectedness and Compactness

3.1 Connected spaces

Definition (Connectedness)

A **separation** of X is a pair of disjoint nonempty open sets with union X. X is **connected** if it has no separation, equivalently if the only sets that are both open and closed are $\emptyset$ and X.

Lemma 3.1

If C, D separate X and $Y \subset X$ is connected, then $Y \subset C$ or $Y \subset D$.

Theorem 3.2

(a) A union of connected sets with a common point is connected. (b) If A is connected and $A \subset B \subset \overline{A}$, then B is connected. (c) Continuous images of connected spaces are connected. (d) Finite products of connected spaces are connected; so is any product in the product topology.

Proof. (a), (b) follow from the lemma. (c) A separation of $f(X)$ pulls back. (d) For $X \times Y$: fix $b \in Y$; for each x, $T_x = (X \times \{b\}) \cup (\{x\} \times Y)$ is connected (two connected sets sharing (x, b)), all T_x contain $X \times \{b\}$, and their union is $X \times Y$; apply (a). For arbitrary products, the points differing from a fixed point a in finitely many coordinates form a connected dense set. ■

$\mathbb{R}^\omega$ with the box topology is not connected: bounded and unbounded sequences separate it.

3.2 Connected subspaces of the real line

A simply ordered set L with more than one element is a **linear continuum** if it has the least upper bound property and between any two elements there is a third.

Theorem 3.3

A linear continuum, and every interval or ray in it, is connected. In particular $\mathbb{R}$ and its intervals are connected.

Proof. Suppose Y convex and A, B separate Y, with $a \in A$, $b \in B$, $a < b$. Let $c = \sup(A \cap [a, b])$. If $c \in B$, then (as B is open in Y) some interval $(d, c] \subset B$, giving a smaller upper bound d. If $c \in A$, then $c < b$ and some $[c, e) \subset A$, so c is not an upper bound. Either way a contradiction. ■

Theorem 3.4 (Intermediate value theorem)

If $f : X \to Y$ is continuous, X connected, Y ordered, and $f(a) < r < f(b)$, then $r = f(c)$ for some c.

X is **path connected** if any two points are joined by a path $[0, 1] \to X$. Path connected spaces are connected; the converse fails:

Example 3.1

(a) The **topologist's sine curve** $\overline{S}$, the closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$, is connected (closure of a connected set) but not path connected: a path from $(0,0)$ into S would, at the first time it leaves $\{0\} \times [-1,1]$, be forced to oscillate.
(b) The **long line** $L = S_\Omega \times [0,1)$ (dictionary order) is a path connected linear continuum which is locally homeomorphic to $\mathbb{R}$ but not imbeddable in $\mathbb{R}^n$.
(c) $\mathbb{R}^n - \{0\}$ $(n \geq 2)$ and S^{n-1} are path connected; $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic (remove a point).

3.3 Components and local connectedness

The **components** of X are the equivalence classes of "lying in a common connected subspace"; they are connected, disjoint, closed, and every connected subspace lies in one. **Path components** are defined with paths. X is **locally (path) connected** if every neighbourhood of every point contains a (path) connected neighbourhood.

Theorem 3.5

X is locally connected iff components of open sets are open. If X is locally path connected, its components and path components coincide.

Proof. For the second statement: path components of a locally path connected space are open; the path component P of x is then also closed (its complement is a union of path components), so it equals the component of x. ■

The topologist's sine curve is connected but not locally connected; $\mathbb{Q}$ has singleton components and is not locally connected.

3.4 Compact spaces

X is **compact** if every open cover has a finite subcover.

Theorem 3.6

(a) Closed subspaces of compact spaces are compact. (b) Compact subspaces of Hausdorff spaces are closed; in fact a compact set and a point outside it have disjoint neighbourhoods. (c) Continuous images of compact spaces are compact. (d) A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Proof. (b) For $x_0 \notin Y$, choose disjoint $U_y \ni y$, $V_y \ni x_0$ for $y \in Y$; finitely many U_y cover Y, and the intersection of the corresponding V_y is a neighbourhood of x_0 missing their union. (d) Closed sets map to compact, hence closed, sets. ■

Lemma 3.7 (Tube lemma)

If Y is compact and N is an open set in $X \times Y$ containing $\{x_0\} \times Y$, then N contains a tube $W \times Y$ with W a neighbourhood of x_0.

Proof. Cover $\{x_0\} \times Y$ by finitely many basis sets $U_i \times V_i \subset N$ and take $W = \bigcap U_i$. ■

Hence finite products of compact spaces are compact (Tychonoff's theorem for infinite products is Chapter 5). X is compact iff every collection of closed sets with the **finite intersection property**

has nonempty intersection.

3.5 Compact subspaces of the real line

Theorem 3.8

In an ordered set with the least upper bound property, closed intervals $[a, b]$ are compact. A subspace of $\mathbb{R}^n$ is compact iff it is closed and bounded (Heine–Borel).

Proof. Given an open cover of $[a, b]$, let C be the set of $y > a$ such that $[a, y]$ is finitely covered. C is nonempty and $c = \sup C$ belongs to some member U of the cover, which contains an interval $(d, c]$; so $c \in C$, and if $c < b$, U also covers points beyond c. Hence $b \in C$. Then $\mathbb{R}^n$: a closed bounded set lies in a product $[-M, M]^n$. ■

Theorem 3.9 (Extreme value theorem)

A continuous $f : X \to Y$, X compact and Y ordered, attains a maximum and a minimum.

Lemma 3.10 (Lebesgue number lemma)

If $\mathcal{A}$ is an open cover of a compact metric space X, there is $\delta > 0$ such that every set of diameter less than δ lies in some member of $\mathcal{A}$.

Proof. Choose a finite subcover $A_1, \ldots, A_n$ (all $\neq X$), let $C_i = X - A_i$ and $f(x) = \frac{1}{n} \sum d(x, C_i)$; $f > 0$ and continuous, so $f \geq \delta > 0$. If x has $d(x, C_m) = \max_i d(x, C_i) \geq f(x) \geq \delta$, the ball $B(x, \delta) \subset A_m$. ■

Consequences: continuous maps from compact metric spaces are uniformly continuous; a nonempty compact Hausdorff space without isolated points is uncountable.

3.6 Limit point compactness and local compactness

X is **limit point compact** if every infinite subset has a limit point, and **sequentially compact** if every sequence has a convergent subsequence.

Theorem 3.11

Compact implies limit point compact. For metrizable spaces, compact, limit point compact and sequentially compact are equivalent.

Proof. If A has no limit point, A is closed and each point has a neighbourhood meeting A in one point; these neighbourhoods with $X - A$ cover X, so A is finite. For metric X, sequential compactness gives a Lebesgue number and total boundedness (X is covered by finitely many ϵ-balls), and these together give compactness. ■

In general these notions differ: S_Ω is limit point compact and sequentially compact but not compact.

X is **locally compact** at x if some compact set contains a neighbourhood of x. $\mathbb{R}^n$ is locally compact; $\mathbb{Q}$ and $\mathbb{R}^\omega$ are not.

Theorem 3.12 (One-point compactification)

A space X is locally compact Hausdorff iff there is a compact Hausdorff Y with $X \subset Y$ and $Y - X$ a single point; Y is unique up to homeomorphism. (Open sets of Y: open sets of X, and $Y - C$ for compact $C \subset X$.)

For example $\mathbb{R}^n \cup \{\infty\} \cong S^n$ by stereographic projection. In a Hausdorff space, local compactness at x is equivalent to: every neighbourhood U of x contains a neighbourhood V with $\overline{V}$ compact and $\overline{V} \subset U$.

Chapter 4. Countability and Separation Axioms

4.1 Countability axioms

X is **first countable** if each point has a countable neighbourhood basis, and **second countable** if the topology has a countable basis. A subset is **dense** if its closure is X; X is **separable** if it has a countable dense subset, and **Lindelöf** if every open cover has a countable subcover.

Theorem 4.1

In a first countable space, $x \in \overline{A}$ iff some sequence in A converges to x, and f is continuous iff it preserves convergent sequences. A second countable space is separable and Lindelöf. Subspaces and countable products of first (second) countable spaces are first (second) countable.

Metric spaces are first countable; for them, second countable, separable and Lindelöf are equivalent. $\mathbb{R}_\ell$ is first countable, separable and Lindelöf but not second countable; $\mathbb{R}_\ell^2$ (the Sorgenfrey plane) is not Lindelöf (the antidiagonal is closed and discrete), so products of Lindelöf spaces need not be Lindelöf.

4.2 Separation axioms

Assume one-point sets are closed. X is **regular** if a point and a disjoint closed set have disjoint neighbourhoods, and **normal** if two disjoint closed sets do. Normal $\Rightarrow$ regular $\Rightarrow$ Hausdorff.

Lemma 4.2

X is regular iff every neighbourhood U of x contains a neighbourhood V of x with $\overline{V} \subset U$; normal iff the same holds with closed sets A in place of points.

Subspaces and products of Hausdorff (regular) spaces are Hausdorff (regular); normality is not preserved: $\mathbb{R}_\ell$ is normal but $\mathbb{R}_\ell^2$ is not, and $S_\Omega \times \overline{S}_\Omega$ is a non-normal subspace of a normal space. $\mathbb{R}_K$ is Hausdorff but not regular.

4.3 Normal spaces

Theorem 4.3

(a) Regular second countable spaces are normal. (b) Metrizable spaces are normal. (c) Compact Hausdorff spaces are normal. (d) Well-ordered sets in the order topology are normal.

Proof. (b) For disjoint closed A, B, take $U = \bigcup_{a\in A} B(a, \epsilon_a/2)$, $V = \bigcup_{b\in B} B(b, \epsilon_b/2)$, where $B(a, \epsilon_a)$ misses B and $B(b, \epsilon_b)$ misses A; the triangle inequality makes $U \cap V = \emptyset$. (c) Apply Theorem 3.6(b) twice. (a) With a countable basis, choose basis sets U_n covering A with closures missing B and V_n covering B with closures missing A; then $U_n' = U_n - \bigcup_{k\le n} \overline{V_k}$ and $V_n' = V_n - \bigcup_{k\le n} \overline{U_k}$ have disjoint unions. ■

4.4 The Urysohn lemma

Theorem 4.4 (Urysohn lemma)

If X is normal and A, B are disjoint closed sets, there is a continuous $f : X \to [0,1]$ with $f = 0$ on A and $f = 1$ on B.

Proof. Enumerate the rationals P in $[0,1]$ with $1, 0$ first. Using Lemma 4.2 recursively, define open sets U_p ($p \in P$) with $A \subset U_0$, $U_1 = X - B$, and $\overline{U_p} \subset U_q$ whenever $p < q$: when U_r is to be defined, it lies between already-defined U_p and U_q with $p < r < q$ adjacent among those defined, and normality gives $\overline{U_p} \subset U_r \subset \overline{U_r} \subset U_q$. Extend by $U_p = \emptyset$ for $p < 0$ and $U_p = X$ for $p > 1$, and put $f(x) = \inf\{p : x \in U_p\}$. If $x \in \overline{U_r}$ then $f(x) \le r$, and if $x \notin U_r$ then $f(x) \ge r$; hence $f^{-1}((c,d))$ is open, as one checks for each x_0 by choosing rationals $c < p < f(x_0) < q < d$ and the neighbourhood $U_q - \overline{U_p}$. ∎

A space where points and closed sets are separated by continuous functions is **completely regular** (Tychonoff). Normal ⇒ completely regular ⇒ regular; subspaces and products of completely regular spaces are completely regular.

4.5 The Urysohn metrization theorem

Theorem 4.5 (Urysohn metrization)

Every regular second countable space is metrizable.

Proof. X is normal (Theorem 4.3(a)). For each pair of basis sets B_n, B_m with $\overline{B_n} \subset B_m$, Urysohn gives $g_{n,m} : X \to [0,1]$ equal to 1 on $\overline{B_n}$ and 0 off B_m. This countable family $\{f_n\}$ separates points from closed sets: for $x \in U$ open, some $f_n(x) > 0$ and $f_n = 0$ off U. Then $F(x) = (f_1(x), f_2(x), \dots)$ imbeds X in $[0,1]^\omega \subset \mathbb{R}^\omega$ (product topology), which is metrizable (Theorem 2.6). Injectivity and continuity are clear; F is open onto its image because $F(U) \supset F(X) \cap \pi_n^{-1}((0,\infty)) \ni F(x)$ for the f_n chosen above. ∎

The proof shows more generally (the **imbedding theorem**): a T_1 space with a family of continuous functions $X \to [0,1]$ separating points from closed sets imbeds in $[0,1]^J$; thus completely regular spaces are exactly the subspaces of compact Hausdorff spaces.

4.6 The Tietze extension theorem

Theorem 4.6 (Tietze extension)

If X is normal and $A \subset X$ is closed, every continuous $f : A \to [a,b]$ (resp. $\mathbb{R}$) extends to a continuous $g : X \to [a,b]$ (resp. $\mathbb{R}$).

Proof. For $f : A \to [-r, r]$, Urysohn gives g with $|g| \le r/3$ on X and $|f - g| \le 2r/3$ on A (separate $f^{-1}([-r, -r/3])$ and $f^{-1}([r/3, r])$). Iterating on the error gives g_n with $|g_n| \le \frac{1}{3}(\frac{2}{3})^{n-1} r$ and $|f - \sum_{k \le n} g_k| \le (\frac{2}{3})^n r$ on A; $\sum g_n$ converges uniformly to an extension. For $\mathbb{R}$-valued f compose with a homeomorphism $\mathbb{R} \cong (-1,1)$ and correct the extension where it reaches ± 1 by multiplying with an Urysohn function that vanishes on the closed set where $|g| = 1$ and equals 1 on A. ∎

Tietze's theorem in turn implies Urysohn's lemma, so it characterizes normality.

4.7 Imbeddings of manifolds

An m-**manifold** is a second countable Hausdorff space in which every point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^m$.

Theorem 4.7

Every compact m-manifold imbeds in $\mathbb{R}^N$ for some N.

Proof. Cover X by finitely many open sets U_i with imbeddings $g_i : U_i \to \mathbb{R}^m$, and choose a **partition of unity** ϕ_i dominated by $\{U_i\}$ (continuous, $\sum \phi_i = 1$, $\operatorname{supp} \phi_i \subset U_i$; it exists in normal spaces by the shrinking lemma and Urysohn). Then $F = (\phi_1, \ldots, \phi_n, \phi_1 g_1, \ldots, \phi_n g_n)$ is an injective continuous map of a compact space into $\mathbb{R}^{n(m+1)}$, hence an imbedding. ■

Chapter 5. The Tychonoff Theorem

5.1 The Tychonoff theorem

Theorem 5.1 (Tychonoff)

An arbitrary product of compact spaces is compact in the product topology.

We use the finite intersection property (FIP) characterization of compactness.

Lemma 5.2

Every collection $\mathcal{A}$ of subsets of X with the FIP is contained in a collection $\mathcal{D}$ that is maximal with respect to the FIP. Such a $\mathcal{D}$ is closed under finite intersections, and any set meeting every member of $\mathcal{D}$ belongs to $\mathcal{D}$.

Proof. Zorn's lemma applied to the collections containing $\mathcal{A}$ with the FIP, ordered by inclusion (the union of a chain still has the FIP, since any finite subfamily lies in one member of the chain). If A meets every member of $\mathcal{D}$, then $\mathcal{D} \cup \{\text{finite intersections of } \mathcal{D}\} \cup \{A\}$ still has the FIP, so by maximality $A \in \mathcal{D}$. ∎

Proof of Tychonoff's theorem. Let $\mathcal{A}$ be a collection of subsets of $X = \prod X_\alpha$ with the FIP; we show $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$. Enlarge $\mathcal{A}$ to a maximal $\mathcal{D}$. For each α, $\{\pi_\alpha(D) : D \in \mathcal{D}\}$ has the FIP in the compact X_α, so pick $x_\alpha \in \bigcap_D \overline{\pi_\alpha(D)}$ and let $x = (x_\alpha)$. For any subbasis set $\pi_\beta^{-1}(U_\beta)$ containing x, U_β meets every $\pi_\beta(D)$, so $\pi_\beta^{-1}(U_\beta)$ meets every D and therefore belongs to $\mathcal{D}$. Finite intersections of these (the basis sets around x) also lie in $\mathcal{D}$ and hence meet every D. So $x \in \overline{D}$ for all $D \in \mathcal{D} \supset \mathcal{A}$. ∎

The theorem is equivalent to the axiom of choice. It fails for the box topology, and for infinite products only the product topology is compact (e.g. $[0,1]^\omega$ is compact, $\mathbb{R}^\omega$ is not since $\mathbb{R}$ is not).

Example 5.1

(a) The Hilbert cube $[0,1]^\omega$ is a compact metrizable space, and by Urysohn's theorem every second countable regular space imbeds in it.
(b) $\{0,1\}^\omega$ is compact and homeomorphic to the Cantor set; $[0,1]^J$ for uncountable J is compact but neither metrizable nor sequentially compact.

5.2 The Stone–Čech compactification

A **compactification** of X is a compact Hausdorff Y containing X as a dense subspace. Only completely regular spaces have compactifications, and every completely regular space has them: imbed X in a cube $[0,1]^J$ and take the closure.

Theorem 5.3

Let X be completely regular. There is a compactification βX such that every bounded continuous $f : X \to \mathbb{R}$ extends uniquely to a continuous function on βX. More generally every continuous map f of X into a compact Hausdorff space C extends uniquely to $\beta X \to C$. βX is unique up to a homeomorphism fixing X.

Proof. Let $\{f_\alpha\}_{\alpha\in J}$ be all bounded continuous real functions on X, with $f_\alpha(X) \subset [a_\alpha, b_\alpha]$. $h = (f_\alpha) : X \to \prod[a_\alpha, b_\alpha]$ is an imbedding (complete regularity), and $\beta X = \overline{h(X)}$ is compact by

Tychonoff. Each f_α extends as the projection π_α. For a compact Hausdorff C, imbed C in a cube $[0,1]^K$; the coordinates of f extend, the extension maps $\beta X = \overline{X}$ into $\overline{f(X)} \subset C$ since C is closed. Uniqueness: extensions agreeing on a dense set agree (Hausdorff target), and the universal property identifies any two such compactifications. ■

Example 5.2

(a) The one-point compactification of $(0,1)$ is a circle; $[0,1]$ is a two-point compactification; $\beta(0,1)$ is much larger, since $\sin(1/x)$ extends.
(b) $\beta\mathbb{Z}_+$ has the cardinality of $\mathcal{P}(\mathcal{P}(\mathbb{Z}_+))$ and is not sequentially compact; it can be described as the space of ultrafilters on $\mathbb{Z}_+$.
(c) $\beta S_\Omega = \overline{S}_\Omega$: every continuous real function on S_Ω is eventually constant.

Chapter 6. Metrization Theorems and Paracompactness

Urysohn's theorem characterizes metrizability among second countable spaces. Here we characterize it in general.

6.1 Local finiteness

A collection $\mathcal{A}$ of subsets of X is **locally finite** if every point has a neighbourhood meeting only finitely many members, and **countably locally finite** (σ-locally finite) if it is a countable union of locally finite collections.

Lemma 6.1

If $\mathcal{A}$ is locally finite, so is $\{\overline{A} : A \in \mathcal{A}\}$, and $\overline{\bigcup A} = \bigcup \overline{A}$.

Proof. An open set meeting $\overline{A}$ meets A. If $x \in \overline{\bigcup A}$, a neighbourhood meeting only $A_1, \dots, A_k$ shows $x \in \overline{A_1 \cup \cdots \cup A_k} = \bigcup_{i \le k} \overline{A_i}$. ∎

A collection $\mathcal{B}$ **refines** $\mathcal{A}$ if every member of $\mathcal{B}$ lies in some member of $\mathcal{A}$.

Lemma 6.2

Every open cover of a metrizable space has a countably locally finite open refinement that covers the space.

Proof. Well-order the cover $\mathcal{A}$. For each n and $U \in \mathcal{A}$ let $S_n(U) = \{x : B(x, 1/n) \subset U\}$, $T_n(U) = S_n(U) - \bigcup_{V<U} V$, and $E_n(U)$ its $1/(3n)$-neighbourhood. For fixed n the sets $E_n(U)$ are at mutual distance at least $1/(3n)$, so $\{E_n(U)\}_U$ is locally finite; and every x lies in some $E_n(U)$, with U the first member containing x and n large. ∎

6.2 The Nagata–Smirnov metrization theorem

Theorem 6.3 (Nagata–Smirnov)

X is metrizable iff it is regular and has a countably locally finite basis.

Sketch. Necessity: apply Lemma 6.2 to the covers by $1/m$-balls, $m = 1, 2, \dots$. Sufficiency: (i) such an X is normal, and every closed set is a G_δ (a countable intersection of open sets), which gives, for each open U, a continuous $f \ge 0$ with $U = \{f > 0\}$. (ii) For each basis element B in the n-th locally finite family $\mathcal{B}_n$ choose $f_{n,B} : X \to [0, 1/n]$ positive exactly on B. (iii) The map $F(x) = (f_{n,B}(x))_{n,B}$ into $[0,1]^J$ with the *uniform* metric is an imbedding: local finiteness makes the uniform metric continuous on $F(X)$ (near a point only finitely many coordinates per n are nonzero, and the factor $1/n$ handles large n), and the separation property makes F open onto its image. ∎

Since a countable basis is countably locally finite, this contains Urysohn's theorem.

6.3 Paracompactness

X is **paracompact** (Hausdorff, by convention) if every open cover has a locally finite open refinement that covers X. Compact spaces and metrizable spaces (Stone's theorem, from Lemma 6.2 with more work) are paracompact; closed subspaces of paracompact spaces are paracompact.

Theorem 6.4

(a) Every paracompact Hausdorff space is normal. (b) Every regular Lindelöf space is paracompact.

Proof. (a) First regularity: for x and closed B, cover X by $X - B$ and neighbourhoods of points of B whose closures miss x; a locally finite refinement $\mathcal{C}$ gives $V = \bigcup\{C \in \mathcal{C} : C \cap B \neq \emptyset\} \supset B$ with $x \notin \overline{V} = \bigcup \overline{C}$ (Lemma 6.1). Normality is the same argument with points replaced by a closed set. (b) A countable cover is refined to a locally finite one by subtracting closures of earlier finite unions. ■

Theorem 6.5 (Partitions of unity)

If X is paracompact Hausdorff and $\{U_\alpha\}$ is an open cover, there is a partition of unity $\{\phi_\alpha\}$ dominated by it: continuous $\phi_\alpha \geq 0$ with $\operatorname{supp} \phi_\alpha \subset U_\alpha$, the supports locally finite, and $\sum \phi_\alpha = 1$.

Proof. Shrink the cover to a locally finite cover $\{V_\alpha\}$ with $\overline{V_\alpha} \subset U_\alpha$ (paracompactness and normality), choose Urysohn functions ψ_α equal to 1 on $\overline{W_\alpha}$ (a further shrinking) and 0 off V_α, and normalize $\phi_\alpha = \psi_\alpha / \sum \psi_\beta$ (a locally finite, positive sum). ■

Partitions of unity let local constructions be glued (Riemannian metrics, imbeddings of manifolds, extensions of maps). $\mathbb{R}_\ell$ is paracompact; $\mathbb{R}_\ell^2$ and S_Ω are not; products of paracompact spaces need not be paracompact.

6.4 The Smirnov metrization theorem

A space is **locally metrizable** if every point has a metrizable neighbourhood.

Theorem 6.6 (Smirnov)

A space is metrizable iff it is paracompact Hausdorff and locally metrizable.

Proof. Cover X by open metrizable sets, take a locally finite open refinement $\mathcal{C}$, and for each C a metric basis of balls in C; for fixed radius $1/m$ the collection of $1/m$-balls in all C refined once more is locally finite. These give a countably locally finite basis; apply Nagata–Smirnov. ■

So a manifold (locally Euclidean, Hausdorff) is metrizable iff it is paracompact; the long line is a non-metrizable 1-manifold-like space because it fails paracompactness.

Chapter 7. Complete Metric Spaces and Function Spaces

7.1 Complete metric spaces

A metric space is **complete** if every Cauchy sequence converges. A Cauchy sequence with a convergent subsequence converges; closed subsets of complete spaces are complete. $\mathbb{R}^k$ is complete (Cauchy sequences are bounded, and Bolzano–Weierstrass applies).

Theorem 7.1

(a) $\mathbb{R}^\omega$ is complete in the metric D of Theorem 2.6. (b) If Y is complete, then Y^J is complete in the uniform metric $\bar{\rho}$. (c) For any space X, the bounded continuous functions $\mathcal{C}(X, Y)$ and the continuous functions with $\bar{\rho}$ form closed subspaces of Y^X, hence complete metric spaces.

Proof. (b) A $\bar{\rho}$-Cauchy sequence is uniformly Cauchy, hence converges coordinatewise, uniformly. (c) Uniform limits of continuous functions are continuous. ■

A metric space can be complete in one metric and not in an equivalent one ($\mathbb{R}$ versus $(-1, 1)$), so completeness is not a topological property.

Theorem 7.2 (Completion)

Every metric space (X, d) isometrically imbeds as a dense subset of a complete metric space $\tilde{X}$, unique up to isometry.

Proof. Fix x_0 and map $x \mapsto \phi_x$, $\phi_x(y) = d(y, x) - d(y, x_0)$, into the complete space of bounded functions with the sup metric; $\|\phi_x - \phi_{x'}\|_\infty = d(x, x')$. Take the closure of the image. ■

Example 7.1 (Peano space-filling curve)

Define piecewise linear paths $f_n : [0, 1] \to [0, 1]^2$ by repeatedly replacing each segment by a zigzag through the four quarter-squares. Then $|f_{n+1} - f_n| \le \sqrt{2}/2^n$ uniformly, so $f_n \to f$ uniformly by completeness of $\mathcal{C}([0, 1], [0, 1]^2)$, and f is continuous; every point of the square is within $\sqrt{2}/2^n$ of $f_n([0, 1])$, so f is onto (its image is compact and dense).

7.2 Compactness in metric spaces

A metric space is **totally bounded** if for every $\epsilon > 0$ it is covered by finitely many ϵ-balls.

Theorem 7.3

A metric space is compact iff it is complete and totally bounded.

Proof. Compact implies these (cover by all ϵ-balls; a Cauchy sequence has a convergent subsequence). Conversely, given a sequence, total boundedness lets us choose, by a diagonal argument, nested subsequences lying in balls of radius $1/k$; the diagonal is Cauchy, hence convergent, so the space is sequentially compact, hence compact (Theorem 3.11). ■

Theorem 7.4 (Ascoli, classical form)

Let X be compact and $\mathcal{F} \subset \mathcal{C}(X, \mathbb{R}^n)$ with the sup metric. Then $\overline{\mathcal{F}}$ is compact iff $\mathcal{F}$ is equicontinuous and pointwise bounded.

Proof. If $\mathcal{F}$ is equicontinuous and pointwise bounded, it is totally bounded: given ϵ, cover the compact X by finitely many δ-neighbourhoods of points $a_1, \dots, a_k$ (from equicontinuity); the values at the a_i lie in a bounded set covered by finitely many small balls, and functions with close values at all a_i are uniformly 3ϵ-close. The closure of an equicontinuous family is equicontinuous, and $\mathcal{C}$ is complete, so $\overline{\mathcal{F}}$ is complete and totally bounded. The converse follows from total boundedness of compact sets. ■

7.3 Pointwise and compact convergence

On Y^X the **topology of pointwise convergence** is the product topology. The **topology of compact convergence** has basis sets $B_C(f, \epsilon) = \{g : \sup_C d(f, g) < \epsilon\}$ for compact $C \subset X$; $f_n \to f$ in it iff $f_n \to f$ uniformly on compact sets. The **compact-open topology** on $\mathcal{C}(X, Y)$ has subbasis $S(C, U) = \{f : f(C) \subset U\}$ (C compact, U open); for metric Y it coincides on $\mathcal{C}(X, Y)$ with the topology of compact convergence, so it depends only on the topology of Y.

Theorem 7.5

If X is locally compact Hausdorff, the evaluation map $e : X \times \mathcal{C}(X, Y) \to Y$, $e(x, f) = f(x)$, is continuous (compact-open topology). Moreover a map $F : X \times Z \to Y$ is continuous iff its adjoint $Z \to \mathcal{C}(X, Y)$, $z \mapsto F(\cdot, z)$, is continuous.

The second statement means that homotopies $X \times I \to Y$ are the same as paths in $\mathcal{C}(X, Y)$ — a viewpoint used in Part II.

7.4 Ascoli's theorem

Theorem 7.6 (Ascoli)

Let X be locally compact Hausdorff and (Y, d) a metric space; give $\mathcal{C}(X, Y)$ the topology of compact convergence. A subset $\mathcal{F}$ has compact closure if $\mathcal{F}$ is equicontinuous under d and $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ has compact closure for each $a \in X$. The converse holds if $Y = \mathbb{R}^n$.

Idea. Imbed $\mathcal{F}$ in the product $\prod_a \overline{\mathcal{F}_a}$, compact by Tychonoff. On equicontinuous families the product topology (pointwise convergence) and the compact convergence topology agree, and the closure of $\mathcal{F}$ in the product consists of continuous functions (equicontinuity passes to pointwise limits). ■

Example 7.2

A sequence of differentiable $f_n : [0, 1] \to \mathbb{R}$ with $|f_n(0)| \le 1$ and $|f_n'| \le 1$ has a uniformly convergent subsequence. Families of holomorphic functions bounded on compact sets are equicontinuous (Cauchy estimates), giving Montel's theorem in complex analysis.

Chapter 8. Baire Spaces and Dimension Theory

8.1 Baire spaces

A set A has **empty interior** if it contains no nonempty open set. X is a **Baire space** if every countable union of closed sets with empty interior has empty interior; equivalently, every countable intersection of dense open sets is dense.

Theorem 8.1 (Baire category theorem)

Every compact Hausdorff space and every complete metric space is a Baire space.

Proof. Let A_n be closed with empty interior and U_0 nonempty open; we find a point of U_0 outside $\bigcup A_n$. Since A_1 has empty interior, $U_0 \not\subset A_1$; choose a nonempty open U_1 with $\overline{U_1} \subset U_0 - A_1$ (regularity), and in the metric case with diameter < 1. Continue: $\overline{U_n} \subset U_{n-1} - A_n$, diameter $< 1/n$. The nested nonempty closed sets $\overline{U_n}$ have a common point (by compactness, or because centres form a Cauchy sequence), which lies in U_0 and in no A_n. ■

Open subspaces of Baire spaces, and locally compact Hausdorff spaces, are Baire. $\mathbb{Q}$ is not Baire (a countable union of points).

Theorem 8.2

If X is Baire and $f_n : X \to Y$ (Y metric) are continuous with $f_n \to f$ pointwise, then the set of points where f is continuous is dense in X.

Idea. For $\epsilon > 0$ the closed sets $A_N = \{x : d(f_n(x), f_m(x)) \le \epsilon \ \forall n, m \ge N\}$ cover X, so the union of their interiors is dense (Baire); on the interior of A_N, f is within 3ϵ of the continuous f_N locally, so its oscillation is at most 3ϵ. Intersect over $\epsilon = 1/k$. ■

For example, derivatives (pointwise limits of difference quotients) are continuous on a dense set; the indicator of $\mathbb{Q}$ is not a pointwise limit of continuous functions.

8.2 A nowhere-differentiable function

Theorem 8.3

In the complete metric space $\mathcal{C}([0,1], \mathbb{R})$ with the sup metric, the functions that are differentiable at some point form a set of first category (a countable union of closed sets with empty interior). In particular continuous nowhere-differentiable functions exist and are dense.

Idea. For n, let F_n be the set of f for which some x has $|f(x+h) - f(x)| \le n|h|$ for all small h (say $0 < |h| \le 1/n$, within $[0,1]$). F_n is closed (compactness of $[0,1]$). It has empty interior: every continuous f is uniformly close to a piecewise linear function, and adding a tiny but very steep sawtooth gives a function outside F_n. Every function differentiable somewhere lies in some F_n; by Baire, their union is not everything. ■

8.3 Introduction to dimension theory

A collection of sets has **order** $m+1$ if some point lies in $m+1$ of its members but none lies in more. A space X has **(covering) dimension** at most m if every open cover has an open refinement of order at most $m+1$ that covers X; $\dim X$ is the least such m.

Theorem 8.4

(a) A closed subspace of X has dimension at most $\dim X$. (b) If $X = Y \cup Z$ with Y, Z closed of finite dimension, $\dim X = \max(\dim Y, \dim Z)$. (c) $[0,1]$ has dimension 1; every compact subspace of $\mathbb{R}^m$ has dimension at most m.

For (c). Cover $\mathbb{R}^m$ by small cubes and perturb their positions so that at most $m+1$ cubes of a suitable family meet at any point (a "brick" arrangement); by a Lebesgue number argument, small enough bricks refine any open cover of a compact set. ■

Points of $\mathbb{R}^N$ are in **general position** if any $k+1 \leq N+1$ of them are affinely independent; general position can be achieved by arbitrarily small perturbations.

Theorem 8.5 (Imbedding theorem)

Every compact metrizable space of dimension m imbeds in $\mathbb{R}^{2m+1}$. In particular every compact m-manifold imbeds in $\mathbb{R}^{2m+1}$.

Idea. In the complete metric space $\mathcal{C}(X, \mathbb{R}^{2m+1})$, the maps that are "$\epsilon$-injective" (points with the same image are within ϵ) form a dense open set: given f, take a fine open cover of order at most $m+1$, a partition of unity ϕ_i, and points z_i in general position near f of the cover sets; $g = \sum \phi_i z_i$ is close to f, and general position (two m-simplices in $\mathbb{R}^{2m+1}$ in general position are disjoint unless they share vertices) makes g ϵ-injective. By Baire, the intersection over $\epsilon = 1/n$ is nonempty, giving an injective map, hence an imbedding (compact domain). ■

Compact m-manifolds have dimension m, and $2m+1$ is sharp in general (some graphs do not imbed in $\mathbb{R}^2$, Chapter 10).